

Karnatak University Dharwad
and
Karnatak University Mathematics Teachers
Association (KUMTA)

B.Sc. II Semester (NEP- Revised)
Laboratory Manual – 2024-25

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Program No. 1: Program to solve examples using Euler's theorem.

(Write a program to verify $b^{\phi(n)} \equiv 1 \pmod{n}$ by Euler's Theorem for given positive integers b and n)

Program:

```
clc;
// Function to compute Euler's Totient Function  $\phi(n)$  by using factor()
function phi=euler_totient(n)
unique_primes = unique(factor(n)); // Get unique prime factors
phi = n;
for i = 1:length(unique_primes)
    phi = phi * (1 - 1 / unique_primes(i)); // Count numbers coprime to n
end
endfunction
// Main program to verify Euler's Theorem
b = 3; // Base
n = 40; // Modulus
if gcd([b, n]) <> 1 then
    disp("b and n are not coprime. Eulers Theorem does not apply.");
else
    phi_n = euler_totient(n); // Compute  $\phi(n)$ 
    disp("b = " + string(b) + ", n = " + string(n) + " and  $\phi(n) =$ " + string(phi_n));
    result = modulo(b^(phi_n), n); // Compute  $a^{\phi(n)} \bmod n$ 
    disp(" Given b and n values, we get  $b^{\phi(n)} \bmod n =$ " + string(result));
    if result == 1 then
        disp("Therefore, by Eulers Theorem, we have:  $b^{\phi(n)} \equiv 1 \pmod{n}$ ");
    else
        disp("Eulers Theorem is NOT satisfied.");
    end
end
end
```

Output:

"b = 3, n = 40 and $\phi(n) = 16$ "

"Given b and n values, we get $b^{\phi(n)} \bmod n = 1$ "

"Therefore, by Eulers Theorem, we have: $b^{\phi(n)} \equiv 1 \pmod{n}$ "

(OR)

```

clc;
// Function to compute Euler's Totient Function  $\phi(n)$ 
function phi=euler_totient(n)
    phi = 0; // Initialize count
    for i = 1:n-1
        if gcd([i, n]) == 1 then
            phi = phi + 1; // Count numbers coprime to n
        end
    end
endfunction
// Main program to verify Euler's Theorem for small values of b & n
// disp("Verifying Eulers Theorem");
b = 2; // Base
n = 105; // Modulus
if gcd([b, n]) <> 1 then
    disp("a and n are not coprime. Eulers Theorem does not apply.");
else
    phi_n = euler_totient(n); // Compute  $\phi(n)$ 
    disp("b = " + string(b) + ", n = " + string(n) + " and  $\phi(n) =$ " + string(phi_n));
    result = modulo(a^(phi_n), n); // Compute  $a^{\phi(n)} \bmod n$ 
    disp(" Given b and n values, we get  $b^{\phi(n)} \bmod n =$ " + string(result));
    if result == 1 then
        disp("Therefore, by Eulers Theorem, we have:  $b^{\phi(n)} \equiv 1 \pmod{n}$ .");
    else
        disp("Eulers Theorem is NOT satisfied.");
    end
end
end

```

(OR)

```
// Function to compute Euler's Totient Function  $\varphi(n)$ 
clc;
function phi=euler_totient(n)
    phi = 0; // Initialize count
    for i = 1:n-1
        if gcd([i, n]) == 1 then
            phi = phi + 1; // Count numbers coprime to n
        end
    end
endfunction
// Modular exponentiation function
function r=mod_exp(b, e, n)
    r = 1;
    b = modulo(b, n);
    while e > 0 do
        if modulo(e, 2) == 1 then
            r = modulo(r * b, n);
        end
        e = floor(e / 2);
        b = modulo(b * b, n);
    end
endfunction
// Main program to verify Euler's Theorem
//disp("Verifying Eulers Theorem");
b = 2; // Base
n = 105; // Modulus
if gcd([a, n]) <> 1 then
    disp("b and n are not coprime. Eulers Theorem does not apply.");
else
    phi_n = euler_totient(n); // Compute  $\varphi(n)$ 
    disp("b = " + string(b) + ", n = " + string(n) + " and  $\phi(n) =$ " + string(phi_n));
    result = mod_exp(b, phi_n, n); // Compute  $a^{\varphi(n)} \bmod n$ 
    disp(" Given a and n values, we get  $b^{\phi(n)} \bmod n =$ " + string(result));
    if result == 1 then
        disp("Therefore, by Eulers Theorem, we have:  $b^{\phi(n)} \equiv 1 \pmod{n}$ .");
    else
        disp("Eulers Theorem is NOT satisfied.");
    end
end
end
```

Examples:

1. Write a program to prove $2^{24} \equiv 1 \pmod{35}$ by Euler's Theorem.
 2. Write a program to prove $2^{48} \equiv 1 \pmod{105}$ by Euler's Theorem.
 3. Write a program to prove $3^{42} \equiv 1 \pmod{49}$ by Euler's Theorem.
 4. Write a program to prove $4^{40} \equiv 1 \pmod{55}$ by Euler's Theorem.
- Write a program to prove $5^{20} \equiv 1 \pmod{33}$ by Euler's Theorem.

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Program No. 2: Program to solve examples using Fermat's theorem.

(Write a program to verify $b^{p-1} \equiv 1 \pmod{p}$ by Fermat's Little Theorem for given positive integer b and prime number p)

Program:

```
// Function to verify Fermat's Little Theorem
```

```
function fermat_little_theorem(b, p)
```

```
    if gcd([b,p]) <> 1 then
```

```
        disp("b and p are not coprime, theorem does not hold");
```

```
    return;
```

```
end
```

```
    result = modulo(b^(p-1), p);
```

```
    if result == 1 then
```

```
        disp("Fermats Little Theorem  $a^{(p-1)} \equiv 1 \pmod{p}$  holds for b = " + string(b) + " and p = " + string(p));
```

```
    else
```

```
        disp("Fermats Little Theorem does NOT hold for b = " + string(b) + " and p = " + string(p));
```

```
    end
```

```
endfunction
```

```
disp("Fermats Little Theorem Verification: ")
```

```
b = 5;
```

```
p = 23;
```

```
fermat_little_theorem(b, p)
```

Output:

```
"Fermats Little Theorem Verification: "
```

```
"Fermats Little Theorem  $b^{(p-1)} \equiv 1 \pmod{p}$  holds for b = 5 and p = 23"
```

(OR)

Write a program to verify $b^{p-1} \equiv 1 \pmod{p}$ by Fermat's Little Theorem for given positive integers b and p .

Program:

// Function to check if a number is prime

function flag=is_prime(n)

 flag = %t; *// Initialize as true (prime)*

if n <= 1 **then**

 disp("Number is not prime (must be greater than 1)");

 flag = %f; *// Return false*

return;

end

for i = 2:int(sqrt(n))

if modulo(n, i) == 0 **then**

 flag = %f; *// Return false if divisible*

return;

end

end

endfunction

// Function to verify Fermat's Little Theorem

function fermat little theorem(b, p)

if ~is_prime(p) **then**

 disp("p = " + string(p) + " is not a prime number, Fermats Little Theorem does not apply.");

return;

end

if gcd([b, p]) <> 1 **then**

 disp("a and p are not coprime, Fermats Little Theorem does not hold.");

return;

end

 result = modulo(b^(p-1), p);

if result == 1 **then**

 disp("Fermats Little Theorem $a^{p-1} \equiv 1 \pmod{p}$ holds for a = " + string(b) + " and p = " + string(p));

else

 disp("Fermats Little Theorem does NOT hold for a = " + string(b) + " and p = " + string(p));

end

endfunction

disp("Fermats Little Theorem Verification: ")

b = 4;

p = 19;

fermat_little_theorem(b, p);

Output:

"Fermats Little Theorem Verification: "

"Fermats Little Theorem: $a^{(p-1)} \equiv 1 \pmod{p}$ holds for $a = 4$ and $p = 19$ "

Examples:

1. Write a program to verify $2^{52} \equiv 1 \pmod{53}$ by Fermat's Little Theorem.
2. Write a program to verify $3^{30} \equiv 1 \pmod{31}$ by Fermat's Little Theorem.
3. Write a program to verify $4^{22} \equiv 1 \pmod{23}$ by Fermat's Little Theorem.
4. Write a program to verify $5^{22} \equiv 1 \pmod{23}$ by Fermat's Little Theorem.

(OR)

Write a program to prove $2^{48} \equiv 1 \pmod{105}$ and verify by Fermat's Little Theorem.

Program:

```
clc;
b=2;
e=48
m=105;
mods = factor(m); // Factors of 105
for i = 1:length(mods)
    disp("Factor " + string(i) + " is: " + string(mods(i)));
    disp("By Fermats Theorem, we get 2^48 mod " + string(mods(i)) + " value " +
string(modulo(2^48, mods(i))));
end
disp("Directy 2^48 mod " + string(m) + " value is " + string(modulo(b^e, m)));
```

Output:

```
" Factor 1 is: 3"
"By Fermats Theorem, we get 2^48 mod 3 value 1"
" Factor 2 is: 5"
"By Fermats Theorem, we get 2^48 mod 5 value 1"
" Factor 3 is: 7"
"By Fermats Theorem, we get 2^48 mod 7 value 1"
"Directy 2^48 mod 105 value is 1"
```

(OR)

```
// Modular exponentiation
clc;
function r=mod_exp(b, e, m)
    r = 1;
    b = modulo(b, m); // Reduce base modulo m
    while e > 0 do
        if modulo(e, 2) == 1 then
            r = modulo(r * b, m); // Multiply and reduce modulo m
        end
        e = floor(e / 2);
        b = modulo(b * b, m); // Square and reduce modulo m
    end
endfunction
// Main program
mods = [3, 5, 7]; // Factors of 105
for i = 1:length(mods)
    disp("By Fermats Theorem on mod " + string(mods(i)) + " is " + string(mod_exp(2, 48,
    mods(i))));
end
disp("Final result on mod 105: " + string(mod_exp(2, 48, 105)));
```

Examples:

1. Write a program to prove $2^{48} \equiv 1 \pmod{35}$ and verify by using Fermat's Little Theorem.
2. Write a program to prove $2^{48} \equiv 1 \pmod{1365}$ and verify by using Fermat's Little Theorem.
3. Write a program to prove $2^{48} \equiv 1 \pmod{1785}$ and verify by using Fermat's Little Theorem.

Write a program to prove $2^{48} \equiv 1 \pmod{173145}$ and verify by using Fermat's Little Theorem.

Program No. 3: Program to solve examples using Wilson's theorem.

(Write a program to verify $(p-1)!+1 \equiv 0 \pmod{p}$ by Wilson's Theorem for a given prime number p)

Program:

```
clc;
// Main program to verify Wilsons Theorem
disp("Wilsons Theorem Verification");
p = 97;
fact_mod_p = modulo(factorial(p-1)+1, p);
disp("((p-1)!+1)(mod p) = " + string(fact_mod_p));
if fact_mod_p == 0 then
    disp("Wilsons Theorem (p-1)!+1 ≡ 0 (mod p) holds for prime number p = " +
        string(p));
else
    disp("Wilsons Theorem is NOT satisfied for p = " + string(p));
end
```

Output:

"Wilsons Theorem Verification"

"((p-1)!+1)(mod p) = 0"

"Wilsons Theorem (p-1)!+1 ≡ 0 (mod p) holds for prime number p = 97"

(OR)

```

// Function to check if a number is prime
clc;
function flag=is_prime(p)
    if p <= 1 then
        flag = %f; // Not prime
        return;
    end
    for i = 2:int(sqrt(p))
        if modulo(p, i) == 0 then
            flag = %f; // Not prime if divisible
            return;
        end
    end
    flag = %t; // Prime if no divisors found
endfunction
// Main program to verify Wilsons Theorem
disp("Wilsons Theorem Verification");
p = 97;
if ~is_prime(p) then
    disp("p = " + string(p) + " is not a prime number. Wilsons Theorem does not apply.");
else
    //fact_val =
    fact_mod_p = modulo(factorial(p-1)+1, p);
    disp("((p-1)!+1)(mod p) = " + string(fact_mod_p));
    if fact_mod_p == 0 then
        disp("Wilsons Theorem (p-1)!+1 ≡ 0 (mod p) holds for prime number p = " + string(p));
    else
        disp("Wilsons Theorem is NOT satisfied for p = " + string(p));
    end
end
end

```

Examples:

1. Write a program to verify $4!+1 \equiv 0 \pmod{5}$ by Wilson's Theorem.
2. Write a program to verify $6!+1 \equiv 0 \pmod{7}$ by Wilson's Little Theorem.
3. Write a program to verify $10!+1 \equiv 0 \pmod{11}$ by Wilson's Little Theorem.

(OR)

Write a program to prove $2(p-3)!+1 \equiv 0 \pmod{p}$ and verify by using Wilson's Theorem for a given prime number p .

Program:

```
clc;
p = 17;
Wilson_fact_mod_p = modulo(factorial(p-1)+1, p);
Require_fact_mod_p = modulo(2*factorial(p-3)+1, p)
disp("((p-1)!+1)(mod p) = " + string(Wilson_fact_mod_p) + " and (2*(p-3)!+1)(mod p) = "
+ string(Require_fact_mod_p));
if Require_fact_mod_p == 0 then
    disp("Therefore, by Wilsons Theorem, we have  $2*(p-3)!+1 \equiv 0 \pmod{p}$  for prime
number p = " + string(p));
else
    disp("Wilsons Theorem can not be used for given p = " + string(p));
end
```

Output:

```
"((p-1)!+1)(mod p) = 0 and (2*(p-3)!+1)(mod p) = 0"
"Therefore, by Wilsons Theorem, we have  $2*(p-3)!+1 \equiv 0 \pmod{p}$  for prime
number p = 17"
"((p-1)!+1)(mod p) = 0"
```

(OR)

```

// Function to check if a number is prime
clc;
function flag=is_prime(p)
    if p <= 1 then
        flag = %f; // Not prime
        return;
    end
    for i = 2:int(sqrt(p))
        if modulo(p, i) == 0 then
            flag = %f; // Not prime if divisible
            return;
        end
    end
    flag = %t; // Prime if no divisors found
endfunction
// Main program to verify Wilsons Theorem
disp("Verification by Wilsons Theorem");
p = 11;
if ~is_prime(p) then
    disp("p = " + string(p) + " is not a prime number. Wilsons Theorem does not apply.");
else
    Wilson_fact_mod_p = modulo(factorial(p-1)+1, p);
    Require_fact_mod_p = modulo(2*factorial(p-3)+1, p) // Compute (p-1)! mod p
    disp("((p-1)!+1)(mod p) = " + string(Wilson_fact_mod_p) + " and (2*(p-3)!+1)(mod p) = "
+ string(Require_fact_mod_p));
    if Wilson_fact_mod_p == Require_fact_mod_p then
        disp("Therefore, by Wilsons Theorem, we have 2*(p-3)!+1 ≡ 0 (mod p) for prime
number p = " + string(p));
    else
        disp("Wilsons Theorem can not be used for given p = " + string(p));
    end
end
end

```

Examples:

1. Write a program to prove $2(4)!+1 \equiv 0 \pmod{7}$ by using Wilson's Theorem.
2. Write a program to prove $2(8)!+1 \equiv 0 \pmod{11}$ by using Wilson's Theorem.
3. Write a program to prove $2(10)!+1 \equiv 0 \pmod{13}$ by using Wilson's Theorem.

Program No. 4: Write a program on reduction formulas to find the value of $\int \sin^n x \, dx$ and $\int \cos^n x \, dx$ for a given x and positive integer n .

Program:

```
// Function for  $\int \sin^n(x) \, dx$  using reduction formula
clc;
function result=integrate_sin(n, x)
    if n == 0 then
        result = x; //  $\int \sin^0(x) \, dx = \int 1 \, dx = x$ 
    elseif n == 1 then
        result = -cos(x); //  $\int \sin(x) \, dx = -\cos(x)$ 
    else
        // Apply reduction formula:  $\int \sin^n(x) \, dx$ 
        result = -sin(x)^(n-1) * cos(x) / n + (n-1) / n * integrate_sin(n-2, x);
    end
endfunction

// Function for  $\int \cos^n(x) \, dx$  using reduction formula
function result=integrate_cos(n, x)
    if n == 0 then
        result = x; //  $\int \cos^0(x) \, dx = \int 1 \, dx = x$ 
    elseif n == 1 then
        result = sin(x); //  $\int \cos(x) \, dx = \sin(x)$ 
    else
        // Apply reduction formula:  $\int \cos^n(x) \, dx$ 
        result = cos(x)^(n-1) * sin(x) / n + (n-1) / n * integrate_cos(n-2, x);
    end
endfunction

n = 3;
x = %pi / 3;
result_sin = integrate_sin(n, x);
result_cos = integrate_cos(n, x);
disp("At n = " + string(n) + " and x = " + string(x));
disp("Integration of sin^"+string(n)+"(x) is " + string(result_sin));
disp("Integration of cos^"+string(n)+"(x) is " + string(result_cos));
```

Output:

"At n = 3 and x = 1.0471976"

"Integration of sin^3(x) is -0.4583333"

"Integration of cos^3(x) is 0.6495191"

Examples:

1. Write a program on reduction formulas to find the value of $\int \sin^n x \, dx$ and $\int \cos^n x \, dx$ for $x = \frac{\pi}{4}$ and $n=3$.
2. Write a program on reduction formulas to find the value of $\int \sin^n x \, dx$ and $\int \cos^n x \, dx$ for $x = \frac{\pi}{3}$ and $n=4$.
3. Write a program on reduction formulas to find the value of $\int \sin^n x \, dx$ and $\int \cos^n x \, dx$ for $x = \frac{\pi}{6}$ and $n=6$.

Reduction Formula

A formula of the form $\int f(x, n) dx = g(x) + \int f(x, k) dx$, where $k < n$ is called a Reduction formula.

Note:

1. Reduction formulae is basically a recurrence relation which reduces integral of functions of higher degree to lower degree.
2. The successive application of the recurrence relation finally end up with a function of degree 0 or 1 so that we can easily complete the integration process.
3. Applications in Fourier analysis, differential equations, and probability theory demonstrated their utility.

Program No. 5(a): Program on Reduction formulas for $\int \sin^n x dx$ (for all positive odd and even integral values of n) with suitable examples

Example 1: Evaluate $\int_0^{\pi/2} \sin^6 x dx$

Solution: Given $I_6 = \int \sin^6 x dx$

We know that reduction formula of

$$I_n = \int \sin^n x dx = \frac{-\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x dx$$

$$\begin{aligned} I_6 = \int \sin^6 x dx &= \frac{-\sin^5 x \cos x}{6} + \frac{5}{6} \int \sin^4 x dx \\ &= \frac{-\sin^5 x \cos x}{6} + \frac{5}{6} \left[-\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \int \sin^2 x dx \right] \\ &= \frac{-\sin^5 x \cos x}{6} - \frac{5}{24} \sin^3 x \cos x + \frac{15}{24} \left[-\frac{\sin x \cos x}{2} + \frac{1}{2} \int \sin^0 x dx \right] \end{aligned}$$

$$I_6 = \int_0^{\pi/2} \sin^6 x dx = \frac{15}{48} \int_0^{\pi/2} 1 dx = \frac{15}{48} [x]_0^{\pi/2} dx$$

$$\begin{aligned} &= \frac{15}{48} \frac{\pi}{2} \\ &= \frac{15}{48} \frac{3.142}{2} \end{aligned}$$

$$I_6 = \int_0^{\pi/2} \sin^6 x dx = 0.4909$$

Exercise:

1. Evaluate $\int_0^{\pi/2} \sin^4 x \, dx$ 2. Evaluate $\int_0^{\pi/2} \sin^8 x \, dx$ 3. Evaluate $\int_0^{\pi/2} \sin^9 x \, dx$

SCILAB Program

```
clc;
```

```
function I_n = sin_reduction(n, x)
```

```
    if n == 0 then
```

```
        I_n = x;
```

```
    elseif n == 1 then
```

```
        I_n = -cos(x);
```

```
    else
```

```
        I_n = ((n - 1) / n) * sin_reduction(n - 2, x) - (1 / n) * cos(x) * (sin(x)^(n - 1));
```

```
    end
```

```
endfunction
```

```
n = 6;
```

```
x = %pi / 2;
```

```
result = sin_reduction(n, x);
```

```
disp("The value of the integral  $\int \sin^n$  + string(n) + "(x) dx at x = " + string(x) + " is:");
```

```
disp(result);
```

```

1
2 clc; // clears all input and out from console.
3 // Define the reduction formula as a function
1 function I_n=sin_reduction(n,x) // (n,x) are arguments
2     if n==0 then
3         I_n=x; // Base case:  $\int 1 \, dx = x$ 
4     elseif n==1 then
5         I_n=-cos(x); // Base case:  $\int \sin(x) \, dx = -\cos(x)$ 
6     else
7         I_n = ((n - 1) / n) * sin_reduction(n - 2, x) - (1 / n) * cos(x) * (sin(x)^(n - 1));
8     end
9 endfunction
13
14 n = 6; // Change this to the desired power of  $\sin^n(x)$ 
15 x = %pi/2; // Evaluate at  $x = \pi/2$ 
16 result = sin_reduction(n, x);
17 disp("The value of the integral  $\int \sin^n + \text{string}(n) + "(x) \, dx$  at  $x = " + \text{string}(x) + "$  is:");
18 disp(result);
19

```

Program No. 5(b): Program on Reduction formulas for $\int \cos^n x dx$ (for all positive odd and even integral values of n) with suitable examples.

Example 1: Evaluate $\int_0^{\pi/2} \cos^5 x dx$

Solution: Given $I_5 = \int \cos^5 x dx$

We know that reduction formula of $I_n = \int \cos^n x dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x dx$

$$\begin{aligned}
 I_5 &= \int \cos^5 x dx \\
 &= \frac{\cos^4 x \sin x}{5} + \frac{4}{5} \int \cos^3 x dx \\
 &= \frac{\cos^4 x \sin x}{5} + \frac{4}{5} \left[\frac{\cos^2 x \sin x}{3} + \frac{2}{3} \int \cos x dx \right] \\
 &= \frac{\cos^4 x \sin x}{5} + \frac{4}{15} \cos^2 x \sin x + \frac{8}{15} \left[\int \cos x dx \right] \\
 I_5 &= \int_0^{\pi/2} \cos^5 x dx = \frac{8}{15} \int_0^{\pi/2} \cos x dx \\
 &= \frac{8}{15} \\
 I_5 &= \int_0^{\pi/2} \cos^5 x dx = 0.5333
 \end{aligned}$$

Exercise:

1. Evaluate $\int_0^{\pi/2} \cos^6 x dx$
2. Evaluate $\int_0^{\pi/2} \cos^7 x dx$
3. Evaluate $\int_0^{\pi/2} \cos^{10} x dx$

SCILAB Program

```

clc;

function I_n = cos_reduction(n, x)

    if n == 0 then
        I_n = x;
    elseif n == 1 then
        I_n = sin(x);
    else
        I_n = ((n - 1) / n) * cos_reduction(n - 2, x) + (1 / n) * sin(x) * (cos(x)^(n - 1));
    end
endfunction

n = 5;
x = %pi / 2;
result = cos_reduction(n, x);
disp("The value of the integral  $\int \cos^n(x) dx$  at  $x =$ " + string(x) + " is:");
disp(result);

```

```

1 clc;
2 // Define the reduction formula as a function
1 function I_n = cos_reduction(n, x)
2     if n == 0 then
3         I_n = x; // Base case:  $\int 1 dx = x$ 
4     elseif n == 1 then
5         I_n = sin(x); // Base case:  $\int \cos(x) dx = \sin(x)$ 
6     else
7         I_n = ((n - 1) / n) * cos_reduction(n - 2, x) + (1 / n) * sin(x) * (cos(x)^(n - 1));
8     end
9 endfunction
12
13 // Example: Compute the integral for a specific n and x
14 n = 5; // Change this to the desired power of  $\cos^n(x)$ 
15 x = %pi / 2; // Evaluate at  $x = \pi/2$ 
16 result = cos_reduction(n, x);
17
18 disp("The value of the integral  $\int \cos^n(x) dx$  at  $x =$ " + string(x) + " is:");
19 disp(result);
20

```

Program No. 6(a): Program on Reduction formulas for $\int \tan^n x dx$ (for all positive odd and even integral values of n) with suitable examples.

Example 1: Evaluate $\int_0^{\pi/4} \tan^8 x dx$

Solution: Given $I_8 = \int \tan^8 x dx$

We know that reduction formula of $I_n = \int \tan^n x dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x dx$

$$\begin{aligned}
 I_8 &= \int \tan^8 x dx \\
 &= \frac{\tan^7 x}{7} - \int \tan^6 x dx \\
 &= \frac{\tan^7 x}{7} - \left[\frac{\tan^5 x}{5} - \int \tan^4 x dx \right] \\
 &= \frac{\tan^7 x}{7} - \frac{\tan^5 x}{5} + \left[\frac{\tan^3 x}{3} - \int \tan^2 x dx \right] \\
 &= \frac{\tan^7 x}{7} - \frac{\tan^5 x}{5} + \frac{\tan^3 x}{3} - \left[\tan x - \int \tan^0 x dx \right] \\
 I_8 &= \int_0^{\pi/4} \tan^8 x dx = \frac{1}{7} - \frac{1}{5} + \frac{1}{3} - 1 + \int_0^{\pi/4} 1 dx \\
 &= \frac{1}{7} - \frac{1}{5} + \frac{1}{3} - 1 + \frac{\pi}{4} \\
 I_8 &= \int_0^{\pi/4} \tan^8 x dx = 0.0616
 \end{aligned}$$

Exercise:

1. Evaluate $\int_0^{\pi/4} \tan^6 x dx$
2. Evaluate $\int_0^{\pi/4} \tan^7 x dx$
3. Evaluate $\int_0^{\pi/4} \tan^4 x dx$

SCILAB Program

```
clc;
function I_n = tan_reduction(n, x)
    if n == 0 then
        I_n = x;
    elseif n == 1 then
        I_n = -log(abs(cos(x)));
    else
        I_n = (1 / (n - 1)) * (tan(x)^(n - 1)) - tan_reduction(n - 2, x);
    end
endfunction
n = 8;
x = %pi / 4;
result = tan_reduction(n, x);

disp("The value of the integral  $\int \tan^n(x) dx$  at  $x =$ " + string(x) + " is:");
disp(result);
```

```

1 clc;
2 // Define the reduction formula as a function
3 function I_n = tan_reduction(n, x)
4     if n == 0 then
5         I_n = x; // Base case:  $\int 1 \cdot dx = x$ 
6     elseif n == 1 then
7         I_n = -log(abs(cos(x))); // Base case:  $\int \tan(x) \cdot dx = -\ln|\cos(x)|$ 
8     else
9         I_n = (1 / (n - 1)) * (tan(x)^(n - 1)) - tan_reduction(n - 2, x);
10    end
11 endfunction
12
13 // Example: Compute the integral for a specific n and x
14 n = 8; // Change this to the desired power of  $\tan^n(x)$ 
15 x = %pi / 4; // Evaluate at  $x = \pi/4$ 
16 result = tan_reduction(n, x);
17
18 disp("The value of the integral  $\int \tan^n + \text{string}(n) + "(x) \cdot dx$  at  $x = \text{string}(x) + "$  is:");
19 disp(result);
20

```

Program No. 6(b): Program on Reduction formulas for $\int \cot^n x dx$ (for all positive odd and even integral values of n) with suitable examples.

Example 1: Evaluate $\int_0^{\pi/4} \cot^6 x dx$

Solution: Given $I_6 = \int \cot^6 x dx$

We know that reduction formula of $I_n = \int \cot^n x dx = -\frac{\cot^{n-1} x}{n-1} - \int \cot^{n-2} x dx$

$$I_6 = \int \cot^6 x dx$$

$$= \frac{\cot^5 x}{5} - \int \cot^4 x dx$$

$$= \frac{\cot^5 x}{5} - \left[-\frac{\cot^3 x}{3} - \int \cot^2 x dx \right]$$

$$= \frac{\cot^5 x}{5} + \frac{\cot^3 x}{3} + \left[-\cot x - \int \cot^0 x dx \right]$$

$$= \frac{\cot^5 x}{5} + \frac{\cot^3 x}{3} - \cot x - \int \cot^0 x dx$$

$$I_6 = \int_0^{\pi/4} \cot^6 x dx = \frac{1}{5} + \frac{1}{3} - 1 - \int_0^{\pi/4} 1 dx$$

$$= \frac{1}{5} + \frac{1}{3} - 1 - \frac{\pi}{4}$$

$$I_6 = \int_0^{\pi/4} \cot^6 x dx = -1.6522$$

Exercise:

1. Evaluate $\int_0^{\pi/4} \cot^5 x dx$

2. Evaluate $\int_0^{\pi/4} \cot^3 x dx$

3. Evaluate $\int_0^{\pi/4} \cot^8 x dx$

SCILAB Program

```

clc;

function I_n = cot_reduction(n, x)

    if n == 0 then
        I_n = x;
    elseif n == 1 then
        I_n = log(abs(sin(x)));
    else
        I_n = -(cotg(x)^(n - 1)) / (n - 1) - cot_reduction(n - 2, x);
    end
endfunction

n = 6;
x = %pi / 4;
result = cot_reduction(n, x);

disp("The value of the integral  $\int \cot^n(x) dx$  at  $x =$ " + string(x) + " is:");
disp(result);

```

```

1 clc;
2 // Define the reduction formula as a function
3 function I_n = cot_reduction(n, x)
4     if n == 0 then
5         I_n = x; // Base case:  $\int 1 dx = x$ 
6     elseif n == 1 then
7         I_n = log(abs(sin(x))); // Base case:  $\int \cot(x) dx = \ln|\sin(x)|$ 
8     else
9         I_n = -(cotg(x)^(n - 1)) / (n - 1) - cot_reduction(n - 2, x);
10    end
11 endfunction
12
13 // Example: Compute the integral for a specific n and x
14 n = 6; // Change this to the desired power of  $\cot^n(x)$ 
15 x = %pi / 4; // Evaluate at  $x = \pi/4$ 
16 result = cot_reduction(n, x);
17
18 disp("The value of the integral  $\int \cot^n(x) dx$  at  $x =$ " + string(x) + " is:");
19 disp(result);
20

```

Program No. 7(a): Program on Reduction formulas for $\int \sec^n x dx$ (for all positive odd and even integral values of n) with suitable examples.

Example 1: Evaluate $\int_0^{\pi/4} \sec^4 x dx$

Solution: Given $I_4 = \int \sec^4 x dx$

We know that reduction formula of $I_n = \int \sec^n x dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x dx$

$$\begin{aligned} I_4 &= \int \sec^4 x dx \\ &= \frac{\sec^2 x \tan x}{3} + \frac{2}{3} \int \sec^2 x dx \\ &= \frac{\sec^2 x \tan x}{3} + \frac{2}{3} \left[\frac{\sec^0 x \tan x}{1} + 0 \right] \\ &= \frac{\sec^2 x \tan x}{3} + \frac{2 \tan x}{3} \end{aligned}$$

$$\begin{aligned} I_4 &= \int_0^{\pi/4} \sec^4 x dx = \frac{(\sqrt{2})^2 (1)}{3} + \frac{2(1)}{3} \\ &= \frac{2}{3} + \frac{2}{3} \\ &= \frac{4}{3} \end{aligned}$$

$$I_4 = \int_0^{\pi/4} \sec^4 x dx = 1.3333$$

Exercise:

1. Evaluate $\int_0^{\pi/4} \sec^7 x dx$

2. Evaluate $\int_0^{\pi/4} \sec^6 x dx$

3. Evaluate $\int_0^{\pi/4} \sec^9 x dx$

SCILAB Program

```

clc;

function I_n = sec_reduction(n, x)

    if n == 0 then

        I_n = x;

    elseif n == 1 then

        I_n = log(abs(sec(x) + tan(x)));

    else

        I_n = (sec(x)^(n - 2) * tan(x)) / (n - 1) + ((n - 2) / (n - 1)) * sec_reduction(n - 2, x);

    end

endfunction

n = 4;

x = %pi / 4;

result = sec_reduction(n, x);

disp("The value of the integral  $\int \sec^n$  " + string(n) + "(x) dx at x = " + string(x) + " is:");

disp(result);

```

```

1 clc;
2 // Define the reduction formula as a function
1 function I_n = sec_reduction(n, x)
2 ... if n == 0 then
3 ...     I_n = x; // Base case:  $\int 1 \cdot dx = x$ 
4 ... elseif n == 1 then
5 ...     I_n = log(abs(sec(x) + tan(x))); // Base case:  $\int \sec(x) \cdot dx = \ln|\sec(x) + \tan(x)|$ 
6 ... else
7 ...     I_n = (sec(x)^(n - 2) * tan(x)) / (n - 1) + ((n - 2) / (n - 1)) * sec_reduction(n - 2, x);
8 ... end
9 endfunction
12
13 // Example: Compute the integral for a specific n and x
14 n = 4; // Change this to the desired power of  $\sec^n(x)$ 
15 x = %pi / 4; // Evaluate at  $x = \pi/4$ 
16 result = sec_reduction(n, x);
17
18 disp("The value of the integral  $\int \sec^n$  " + string(n) + "(x) dx at x = " + string(x) + " is:");
19 disp(result);
20

```

Program No. 7(b): Program on Reduction formulas for $\int \csc^n x dx$ (for all positive odd and even integral values of n) with suitable examples.

Example 1: Evaluate $\int_0^{\pi/4} \csc^6 x dx$

Solution: Given $I_6 = \int \csc^6 x dx$

We know that reduction formula of $I_n = \int \csc^n x dx = \frac{-\csc^{n-2} x \cot x}{n-1} + \frac{n-2}{n-1} \int \csc^{n-2} x dx$

$$\begin{aligned} I_6 &= \int \csc^6 x dx \\ &= \frac{-\csc^4 x \cot x}{5} + \frac{4}{5} \int \csc^4 x dx \\ &= \frac{-\csc^4 x \cot x}{5} + \frac{4}{5} \left[\frac{-\csc^2 x \cot x}{3} + \frac{2}{3} \int \csc^2 x dx \right] \\ &= \frac{-\csc^4 x \cot x}{5} - \frac{4}{15} \csc^2 x \cot x + \frac{8}{15} \int \csc^2 x dx \\ &= \frac{-\csc^4 x \cot x}{5} - \frac{4}{15} \csc^2 x \cot x + \frac{8}{15} (-\cot x) \\ I_6 &= \int_0^{\pi/4} \csc^6 x dx = \frac{-(\sqrt{2})^4(1)}{5} - \frac{4(\sqrt{2})^2(1)}{15} - \frac{8}{15}(1) \\ &= -\frac{4}{5} - \frac{8}{15} - \frac{8}{15} \end{aligned}$$

$$I_6 = \int_0^{\pi/4} \csc^6 x dx = -1.8666$$

Exercise:

1. Evaluate $\int_0^{\pi/4} \csc^8 x dx$

2. Evaluate $\int_0^{\pi/4} \csc^5 x dx$

3. Evaluate $\int_0^{\pi/4} \csc^{10} x dx$

SCILAB Program

```

clc;

function I_n = csc_reduction(n, x)

    if n == 0 then
        I_n = x;
    elseif n == 1 then
        I_n = -log(abs(csc(x) + cotg(x)));
    else
        I_n = -(csc(x)^(n - 2) * cotg(x)) / (n - 1) + ((n - 2) / (n - 1)) * csc_reduction(n - 2, x);
    end
endfunction

n = 4;
x = %pi / 4;
result = csc_reduction(n, x);

disp("The value of the integral  $\int \csc^n(x) dx$  at  $x =$ " + string(x) + " is:");
disp(result);

```

```

1 clc;
2 // Define the reduction formula as a function
1 function I_n = csc_reduction(n, x)
2     if n == 0 then
3         I_n = x; // Base case:  $\int 1 dx = x$ 
4     elseif n == 1 then
5         I_n = -log(abs(csc(x) + cotg(x))); // Base case:  $\int \csc(x) dx = -\ln|\csc(x) + \cot(x)|$ 
6     else
7         I_n = -(csc(x)^(n - 2) * cotg(x)) / (n - 1) + ((n - 2) / (n - 1)) * csc_reduction(n - 2, x);
8     end
9 endfunction
12
13 // Example: Compute the integral for a specific n and x
14 n = 6; // Change this to the desired power of  $\csc^n(x)$ 
15 x = %pi / 4; // Evaluate at  $x = \pi/4$ 
16 result = csc_reduction(n, x);
17
18 disp("The value of the integral  $\int \csc^n(x) dx$  at  $x =$ " + string(x) + " is:");
19 disp(result);
20

```

Line 13, Column 55.

Program No. 8(a): Program on Reduction formulas for $\int x^n e^{ax} dx$ (for all positive odd and even integral values of n) with suitable examples.

Example 1: Evaluate $\int x^2 e^{3x} dx$

Solution: Given $I = \int x^2 e^{3x} dx$

We know that reduction formula of $I = \int x^n e^{ax} dx = \frac{x^n e^{ax}}{a} - \frac{n}{a} \int x^{n-1} e^{ax} dx$

$$\begin{aligned} I &= \int x^2 e^{3x} dx \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \int x e^{3x} dx \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left[\frac{x e^{3x}}{3} - \frac{1}{3} \int x^0 e^{3x} dx \right] \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{9} x e^{3x} + \frac{2}{9} \int e^{3x} dx \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} \\ I &= \int x^2 e^{3x} dx \Big|_{at\ x=0} = 0.07407 \end{aligned}$$

Exercise:

1. Evaluate $\int x^4 e^{5x} dx$ at $x=2$
2. Evaluate $\int x^5 e^x dx$ at $x=1$
3. Evaluate $\int x^3 e^x dx$ at $x=3$

SCILAB Program

```

function result = reduction_xn_ex(n, a, x)

    if n == 0 then
        result = exp(a * x) / a;
    else
        term1 = (x^n * exp(a * x)) / a;
        term2 = (n / a) * reduction_xn_ex(n - 1, a, x);
        result = term1 - term2;
    end
endfunction

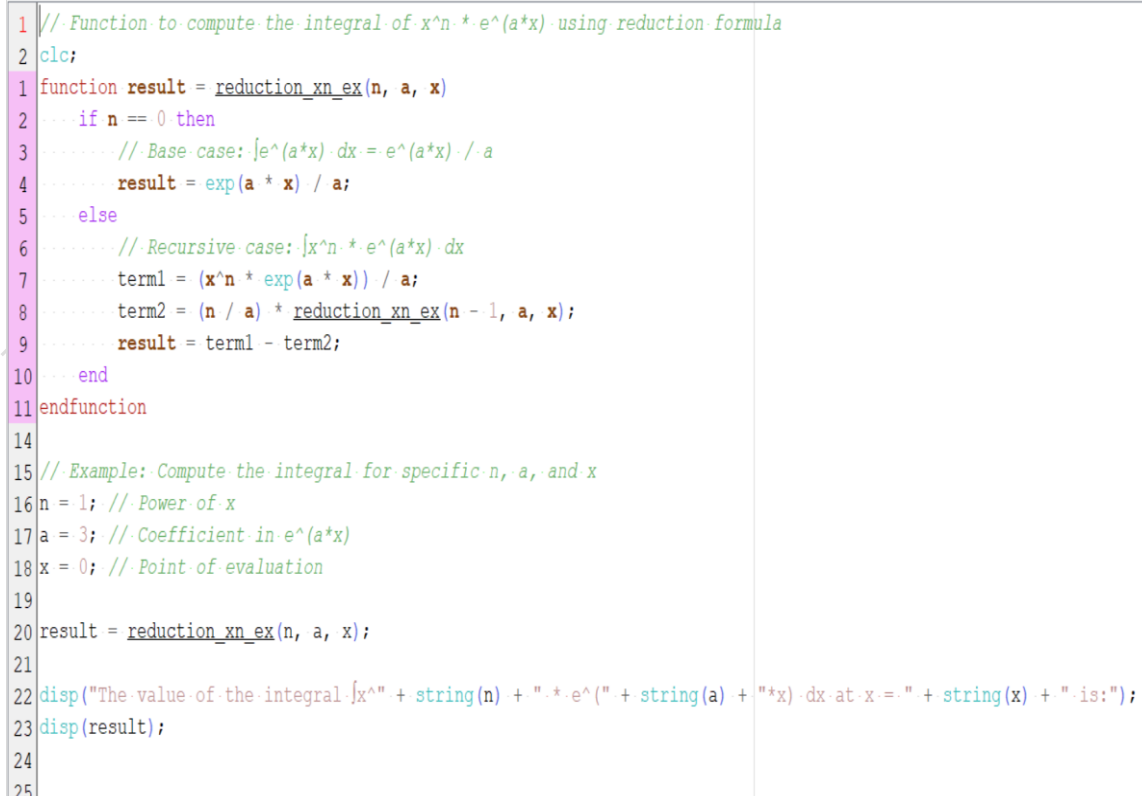
n = 3;
a = 2;
x = 1;

result = reduction_xn_ex(n, a, x);

disp("The value of the integral  $\int x^n * e^{ax} dx$  at  $x =$  " +
string(x) + " is:");

disp(result);

```



```

1 //Function to compute the integral of x^n * e^(ax) using reduction formula
2 clc;
3 function result = reduction_xn_ex(n, a, x)
4     if n == 0 then
5         //Base case:  $\int e^{ax} \cdot dx = e^{ax} / a$ 
6         result = exp(a * x) / a;
7     else
8         //Recursive case:  $\int x^n \cdot e^{ax} \cdot dx$ 
9         term1 = (x^n * exp(a * x)) / a;
10        term2 = (n / a) * reduction_xn_ex(n - 1, a, x);
11        result = term1 - term2;
12    end
13 endfunction
14
15 // Example: Compute the integral for specific n, a, and x
16 n = 1; // Power of x
17 a = 3; // Coefficient in e^(ax)
18 x = 0; // Point of evaluation
19
20 result = reduction_xn_ex(n, a, x);
21
22 disp("The value of the integral  $\int x^n * e^{ax} dx$  at  $x =$  " +
23 string(x) + " is:");
24 disp(result);
25

```

Program No. 8(b): Program on Reduction formulas for $\int x^m (\log x)^n dx$ (for all positive odd and even integral values of n, m) with suitable examples.

Example 1: Evaluate $\int x^3 (\log x)^2 dx$

Solution: Given $I = \int x^3 (\log x)^2 dx$

We know that reduction formula of $I = \int x^m (\log x)^n dx = \frac{x^{m+1} (\log x)^n}{m+1} - \frac{n}{m+1} \int x^m (\log x)^{n-1} dx$

$$\begin{aligned} I &= \int x^3 (\log x)^2 dx \\ &= \frac{x^4 (\log x)^2}{4} - \frac{2}{4} \int x^3 (\log x)^1 dx \\ &= \frac{x^4 (\log x)^2}{4} - \frac{2}{4} \left[\frac{x^4 (\log x)}{4} - \frac{1}{4} \int x^3 dx \right] \\ &= \frac{x^4 (\log x)^2}{4} - \frac{2}{16} x^4 (\log x) + \frac{2}{16} \frac{x^4}{4} \end{aligned}$$

$$I = \int x^3 (\log x)^2 \Big|_{at x=1} = 0.03125$$

Exercise:

1. Evaluate $\int x^4 (\log x) dx$ at $x = 3$
2. Evaluate $\int x^2 (\log x)^2 dx$ at $x = 1$
3. Evaluate $\int x^3 (\log x)^3 dx$ at $x = 0$

SCILAB Program

```

function result = reduction_xm_logxn(m, n, x)

    if n == 0 then
        result = x^(m + 1) / (m + 1);
    else
        term1 = (x^(m + 1) * log(x)^n) / (m + 1);
        term2 = (n / (m + 1)) * reduction_xm_logxn(m, n - 1, x);
        result = term1 - term2;
    end
endfunction

m = 2;
n = 3;
x = 2;

result = reduction_xm_logxn(m, n, x);

disp("The value of the integral  $\int x^m * \log(x)^n dx$  at x = " +
string(x) + " is:");

disp(result);

```

```

1 //Function to compute the integral of  $x^m * \log(x)^n$  using reduction formula
2 clc;
3 function result = reduction_xm_logxn(m, n, x)
4 ... if n == 0 then
5 ...     //Base case:  $\int x^m dx = x^{m+1} / (m+1)$ 
6 ...     result = x^(m + 1) / (m + 1);
7 ... else
8 ...     //Recursive case:  $\int x^m * \log(x)^n dx$ 
9 ...     term1 = (x^(m + 1) * log(x)^n) / (m + 1);
10 ...    term2 = (n / (m + 1)) * reduction_xm_logxn(m, n - 1, x);
11 ...    result = term1 - term2;
12 ... end
13 endfunction
14
15 // Example: Compute the integral for specific m, n, and x
16 m = 3; // Power of x
17 n = 2; // Power of log(x)
18 x = 1; // Point of evaluation
19
20 result = reduction_xm_logxn(m, n, x);
21
22 disp("The value of the integral  $\int x^m * \log(x)^n dx$  at x = " + string(x) + " is:");
23 disp(result);
24
25

```

Program No. 9(a): Program on Reduction formulas for $\int \sin^m x \cos^n x dx$ (for all positive values of m and n) with suitable examples

*//Program on reduction formulas for $\int \sin^m(x) \cos^n(x) dx$ (for all positive values of n)
//by decomposition of sine function*

clc

function result=sin_cos_reduction(m, n)

if n==0 **then**

if m==0 **then**

 result = "x"

elseif m == 1 **then**

 result = "(-cos(x))"

end

elseif m==0 **then**

if n==1 **then**

 result="sin(x)"

else

 s=sin_cos_reduction(0,n-2)

 result = "(1/" + string(n) + ")sin(x) cos(x)^(n-1) + (" + string(n-1) + "/" + string(n-1) + "/" + string(n) + ")(" + s + ")"

end

elseif m>1 **then**

 s=sin_cos_reduction(m-2,n)

 result = "(-1/" + string(m+n) + ")(sin(x)^(m-1) + cos(x)^(n+1) + string(n+1) + " + (" + string(m-1) + "/" + string(m+n) + ")(" + s + ")")"

else

 result = "(-1/" + string(m+n) + "cos(x)^(n+1) + string(n+1)

end

endfunction

m=input("Enter the sine power m ")

n=input("Enter the cosine power n ")

res=sin_cos_reduction(m,n)

disp("The value of integral is")

disp(res)

Output

Enter the sine power m 3

Enter the cosine power n 2

The value of integral is

$(-1/5)(\sin(x)^2 \cos(x)^3) + (2/5)((-1/3)\cos(x)^3)$

Program No. 9(b): Program on Reduction formulas for $\int_0^{\pi/2} \sin^m x \cos^n x dx$ (for all positive values of m and n) with suitable examples.

Example 1: Evaluate $I = \int_0^{\pi/2} \sin^4 x \cos^2 x dx$

Solution: Given $I = \int \sin^4 x \cos^2 x dx$

We know that reduction formula

$$I = \int \sin^m x \cos^n x dx = \frac{-\cos^{n+1} x \sin^{m-1} x}{m+n} + \frac{m-1}{m+n} \int \sin^{m-2} x \cos^n x dx$$

$$\begin{aligned} I &= \int \sin^4 x \cos^2 x dx = \frac{-\cos^3 x \sin^3 x}{6} + \frac{3}{6} \int \sin^2 x \cos^2 x dx \\ &= \frac{-\cos^3 x \sin^3 x}{6} + \frac{1}{2} \left[\frac{-\cos^3 x \sin x}{4} + \frac{1}{4} \int \sin^0 x \cos^2 x dx \right] \\ &= \frac{-\cos^3 x \sin^3 x}{6} - \frac{\cos^3 x \sin x}{8} + \frac{1}{8} \int \cos^2 x dx \\ &= \frac{-\cos^3 x \sin^3 x}{6} - \frac{\cos^3 x \sin x}{8} + \frac{1}{8} \left[\frac{\cos x \sin x}{2} + \frac{1}{2} \int \cos^0 x dx \right] \\ &= \frac{-\cos^3 x \sin^3 x}{6} - \frac{\cos^3 x \sin x}{8} + \frac{\cos x \sin x}{16} + \frac{1}{16} \int 1 dx \end{aligned}$$

$$I = \int_0^{\pi/2} \sin^4 x \cos^2 x dx = \frac{1}{16} \int_0^{\pi/2} 1 dx = \frac{1}{16} [x]_0^{\pi/2} = \frac{1}{16} \frac{\pi}{2} = \frac{\pi}{32}$$

$$I = \int_0^{\pi/2} \sin^4 x \cos^2 x dx = 0.09818$$

Exercise:

1. Evaluate $I = \int_0^{\pi/2} \sin^2 x \cos^4 x dx$
2. Evaluate $I = \int_0^{\pi/2} \sin^2 x \cos^2 x dx$
3. Evaluate $I = \int_0^{\pi/2} \sin^3 x \cos^4 x dx$

SCILAB Program

```

clc;
function result=reduction_sinm_cosn(m, n, x)
    if m == 0 then
        if n == 0 then
            result = x;
        else
            result = sin(x)*cos(x)^(n-1) / n + (n-1)/n *reduction_sinm_cosn(0, n - 2, x);
        end
    elseif n == 0 then
        if m == 1 then
            result = -cos(x);
        else
            result = -cos(x) * sin(x)^(m - 1) / m + (m - 1) / m * reduction_sinm_cosn(m - 2, 0,
x);
        end
    else
        result = (-sin(x)^(m - 1) * cos(x)^(n + 1) / (m+n)) + ((m - 1) / (m+n)) *
reduction_sinm_cosn(m - 2, n, x);
    end
endfunction

m = 4;
n = 2;
x = %pi / 2;

result = reduction_sinm_cosn(m, n, x);

disp("The value of the integral  $\int \sin^m(x) \cos^n(x) dx$ 
at x = " + string(x) + " is:");
disp(result);

```

```

1 clc;
2 %%Function to compute the integral of sin^m(x) * cos^n(x) using reduction formula
1 function result=reduction_sinn_cossn(m, n, x)
2 %%if m == 0 then
3 %%Base case: [cos^n(x) * dx
4 if n == 0 then
5 result = x; %%[1 * dx = x
6 else
7 result = sin(x) * cos(x)^(n-1) / n + (n-1)/n * reduction_sinn_cossn(0, n-2, x); %%Recursive for cos^n(x)
8 end
9 elseif n == 0 then
10 %%Base case: [sin^m(x) * dx
11 if m == 1 then
12 result = -cos(x); %%[sin(x) * dx = -cos(x)
13 else
14 result = -cos(x) * sin(x)^(m-1) / m + (m-1)/m * reduction_sinn_cossn(m-2, 0, x); %%Recursive for sin^m(x)
15 end
16 else
17 %%General case using integration by parts
18 result = (-sin(x)^(m-1) * cos(x)^(n+1) / (m+n)) + ((m-1)/m) * reduction_sinn_cossn(m-2, n, x);
19 end
20 endfunction
23
24 %%Example: Compute the integral for specific m, n, and x
25 m = 4; %%Power of sin(x)
26 n = 2; %%Power of cos(x)
27 x = pi/2; %%Evaluate at x = pi/2
28
29 result = reduction_sinn_cossn(m, n, x);
30
31 disp('The value of the integral [sin^' + string(m) + '(x) * cos^' + string(n) + '(x) * dx at x = ' + string(x) + ' is:');
32 disp(result);
33

```

Program No. 10: Program to verify the Euler's theorem and its extension*//Program to verify the Euler's theorem and its extension*

```

clc
function u=f(x)
u=(x(1)^3+x(2)^3)/(x(1)+x(2));
endfunction
n=2 //order of homogeneous function
x=[1,1]
[fu,su]=numderivative(f,x)
disp("First Partial derivative of u (ux and uy) are")
disp(fu)
disp("second Partial derivative of u (uxx,uxy,uyx and uyy) are")
disp(su)
L=x(1)*fu(1)+x(2)*fu(2)
R=n*f(x)
disp("x ux + y uy = "+string(L)+" and nu = "+string(R))
if abs(R-L) <= 0.00001 then
disp("Eulers Theorem is verified")
else
disp("Eulers Theorem is not verified")
end
L=x(1)^2*su(1)+2*x(1)*x(2)*su(2)+x(2)^2*su(4)
R=n*(n-1)*f(x)
disp("x^2 uxx+2xy uxy+y^2 uyy = "+string(L)+" and n(n-1)u = "+string(R))
if abs(R-L) <=0.00001 then
disp("Eulers Extension Theorem is verified")
else
disp("Eulers extension Theorem is not verified")
end

```

Output:

First Partial derivative of u (ux and uy) are

1. 1.

Second Partial derivative of u (uxx,uxy,uyx and uyy) are

2. -1. -1. 2.

x ux + y uy = 2 and nu = 2

Eulers Theorem is verified

x^2 uxx+2xy uxy+y^2 uyy = 2 and n(n-1)u = 2

Eulers Extension Theorem is verified

Program No. 11: Programs to construct series using Maclaurin's expansion for functions of two variables

//Programs to construct series using Maclaurin's expansion for functions of two variables

```

clc
function u=f(p)
u=exp(p(1))*cos(p(2))
endfunction
p=[0,0]
[fu,su]=numderivative(f,p)
disp("First Partial derivative of u->ux and uy are")
disp(fu)
disp("second Partial derivative of u->uxx,uxy,uyx and uyy are")
disp(su)
t=""
if f(p) ~= 0 then
    t=string(f(p))+ "+"
end
if fu(1)~=0 then
    t=t+"("+string(fu(1))+")x+"
end
if fu(2)~=0 then
    t=t+"("+string(fu(2))+")y+"
end
if su(1)~=0 then
    t=t+"("+string(su(1))+"/2)x^2+"
end
if su(2)~=0 then
    t=t+"("+string(su(2))+")xy+"
end
if su(4)~=0 then
    t=t+"("+string(su(4))+"/2)y^2+"
end
disp("the expansion of function using Maclaurin series upto 2nd degree is ")
disp("e^x cos(y) = "+t+"...")

```

Output:

First Partial derivative of u->ux and uy are

1. 0.

second Partial derivative of u->uxx,uxy,uyx and uyy are

1. 0. 0. -1.

the expansion of function using Maclaurin series upto 2nd degree is

$e^x \cos(y) = 1 + (1)x + (1/2)x^2 + (-1/2)y^2 + \dots$

Program No. 12: Program to find gradient, divergence and curl**Gradient of Scalar Function***//Program to find gradient*

clc

function F=fun(x)

F=x(1)^2*x(2)^2*x(3)^2

endfunction

disp("Scalar function F= x^2*y^2*z^2")

x=[1,1,2]

f=numderivative(fun,x)

disp("grad of F is "+string(f(1))+ "i"+string(f(2))+ "j"+string(f(3))+ "k")

OUTPUT

Scalar function F= x^2*y^2*z^2

grad of F is 8i+8j+4k

Divergence of the vector function*//Program to find divergence*

clc

function F=fun(x)

F(1)=x(1)^2*x(2)

F(2)=x(2)*x(1)

F(3)=x(2)^2*x(3)

endfunction

disp("Vector function F=x^2y i+xy j+x^2z k")

x=[-1,2,3]

f=numderivative(fun,x)disp("Partial derivatives computation are $\partial F1/\partial x, \partial F1/\partial y, \partial F1/\partial z$ etc")

disp(f)

//div F is $\partial F1/\partial x + \partial F2/\partial y + \partial F3/\partial z$

df=f(1,1)+f(2,2)+f(3,3)

disp("div of F="+string(df))

OUTPUT

Vector function F=x^2y i+xy j+x^2z k

Partial derivatives computation are $\partial F1/\partial x, \partial F1/\partial y, \partial F1/\partial z$ etc

- 4. 1. 0.

2. - 1. 0.

0. 12. 4.

div of F=-1

Curl of vector function

```
//Program to find curl
clc
function F=fun(x)
F(1)=x(1)^2*x(2)
F(2)=x(2)*x(1)
F(3)=x(2)^2*x(3)
endfunction
disp("Vector function F=x^2y i+xy j+x^2z k")
x=[-1,2,3]
f=numderivative(fun,x)
disp("Partial derivatives computation are  $\partial F_1/\partial x, \partial F_1/\partial y, \partial F_1/\partial z$  etc")
disp(f)
//curl F is  $(\partial F_3/\partial y - \partial F_2/\partial z)i + (\partial F_1/\partial z - \partial F_3/\partial x)j + (\partial F_2/\partial x - \partial F_1/\partial y)k$ 
cf=[f(3,2)-f(2,3),f(1,3)-f(3,1),f(2,1)-f(1,2)]
disp("curl of F="+string(cf(1))+ "i+" +string(cf(2))+ "j+" +string(cf(3))+ "k")
```

OUTPUT

Vector function $F=x^2y \mathbf{i}+xy \mathbf{j}+x^2z \mathbf{k}$

Partial derivatives computation are $\partial F_1/\partial x, \partial F_1/\partial y, \partial F_1/\partial z$ etc

- 4. 1. 0.

2. - 1. 0.

0. 12. 4.

curl of $F=12\mathbf{i}+0\mathbf{j}+1\mathbf{k}$